

Ryan Shin

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EDUCATION

Duke University | **GPA: 4.0/4.0** | Duke IEEE Exec Member, Taekwondo Club, Duke AERO | **Aug. 2025 - May 2029**
B.S.E. in Electrical and Computer Engineering + Computer Science Durham, NC

- **Relevant Work:** Fields & Waves, Computer Architecture, Signals & Systems, Microelectronics, Humanoid Robotics

Georgia Institute of Technology | **GPA: 4.0/4.0** | Dual Enrollment **Aug. 2023 - May 2025**

PUBLICATIONS

Xia, B., Yang, Z., **Shin, R.**, et al. Visible-Reachable Workspace for Perception-Aware Humanoid Design. arXiv:2609.08905, 2026. [arXiv]

WORK EXPERIENCE

General Robotics Lab | *Humanoid Robot Researcher* (Durham, NC) | **Jan 2026 – Present**

- Built and integrated the electrical stack for a 31-DoF humanoid, including 48-V LiPo power distribution, differential CAN networking across 20+ actuators, centralized power management, and TVS overvoltage protection for reliable high-current operation.
- Developed low-level motor infrastructure by configuring RobStride QDD motors, CAN communication, calibration utilities in Linux, and a relay-based torque testing platform to characterize actuator performance and reliability.
- Designed a specialized parallel gripper with IPC socket-based control and MuJoCo validation workflows, then closed the loop with a visual-feedback MPC controller, cuRobo trajectory planner, and SAC locomotion policy.

Duke ECE Department | *Software Developer* (Durham, NC) | **May 2026 – August 2026**

- Developed full-stack features for the P4ML automated course management platform, streamlining assignment delivery, grading, and repository management for instructors and students using Git-based workflows and MVC architecture.
- Designed relational database schemas and optimized SQL/JDBC queries supporting users, assessments, and grading workflows; build services with Spring Boot, Docker, and Kubernetes.

Georgia Tech CRAB Lab | *Lab Research Member* (Atlanta, GA) | **May 2024 - August 2024**

- Designed sensor PCBs in KiCad and assembled, soldered, and wired Arduino-based, biomechanical force bots
- Designed 3D CAD components to reduce friction and improve mobility, iterating hardware design based on lab feedback.

Caterpillar Switchgear LLC | *Engineering Department Intern* (Alpharetta, GA) | **May 2024 - August 2024**

- Worked with electrical engineers to understand power systems, switchgear architecture, and manufacturing needs
- Improved documentation workflow via automated MS VBA tools, decreasing formatting time and improving interface

Georgia Tech Research Institute | *SWE Intern Researcher* (Atlanta, GA) | **June 2023 - July 2023**

- Developed a full-stack web application using Flask, HTML/CSS, and MySQL for user progress and login
- Visualized user progress via learning modules and implemented HuggingFace API for user question-response queries

PROJECTS

Robodog Quadruped Bot | Designed, built, and integrated an 8-DOF quadruped robot using MuJoCo environment featuring PPO policy with custom gait generation, reward shaping, and trajectory optimization + PD-controlled actuators, domain randomization, IMU and foot-contact sensing, and Sim2Real compatible dynamics.

Cartesian Gripper | Contributed to the design and testing of a 7-DoF Cartesian Hand; designed electronics, developed firmware and contact-aware joint control, and deployed the system on Duke Humanoid V2 and Franka FR3 robot arm respectively for bimanual manipulation and sequenced task execution.

Compost Bin Alert System | Developed full-stack IoT compost monitoring prototype for CompostNow using ESP32 and ultrasonic sensors to track fill levels with live updating via Firebase.

Autonomously Regulating HABs by Mobilizing UV-Pump Filtration | Developed a cost-effective and accurate autonomous detection system for algae concentration detection and near 100% UV removal using turbidity, TDS, color sensors. Reviewed over 25+ peer-reviewed research sources.

SKILLS + AWARDS

Skills: Microcontrollers, Java, Python (PyTorch), Fusion360, CAN, C++, Docker, SQL, Linux, KiCad, MuJoCo.
Awards: Duke Dean's List

WEBSITE: <https://ryan-shin14.github.io/index.html>
